1

Instrumentation for food quality assurance

E. Kress-Rogers, ALSTOM, Ratingen

1.1 Introduction

1.1.1 The role of quality assurance in the food industry

Quality control is essential in the food industry, and efficient quality assurance is becoming increasingly important. The consumer expects a wide range of competitively priced food products of consistently high quality. Each food item has to be safe, wholesome and attractive in appearance, taste and texture, and needs to be consistent with the product image. Variations within the same batch or between batches will have to be kept to a minimum as they are often interpreted by the consumer as indicating a fault, even when the differing product is of high quality.

The availability, quality and price of raw materials will place conditions on the food manufacturing operation, as will the prevailing structure of the retailing sector. More and more frequently, the product palette has to be adapted to changes in tastes and nutritional ideas, and to the appearance of competing products on the market. In the manufacture of each new product, there is the challenge of getting it right first time. Increasingly, food processing operations are technology-based rather than skill-based. Legislation on food composition and labelling will also play a role. Changes in legislation are driven by consumer demands and by international harmonization.

Food processing has a long history (Georgala 1989) and has always had two main purposes. The first is the conversion of agricultural products (or of fished, hunted and gathered foods) into palatable, attractive, digestible and safe foods. Cereals, for example, are virtually inedible without prior milling and cooking or fermenting; some fruits and pulses are toxic without prior cooking; and large proportions of the Asian and African populations can consume lactose only after conversion to lactate by fermentation. The second purpose is the preservation of foods for availability out of season, for years of lean harvests, and for transport to areas distant from agricultural producers.

The assessment of food still centres on its taste, aroma, appearance and nutritional value, and on its safety and stability. Optimized process control plays an essential part in maintaining the commercial viability of a food manufacturing operation in the face of changes in the food market and in the structure of the food industry. Advances in

Table 1.1 Food and drink market sectors in the UK (Bailey *et al.* 1991)

(a) Market sectors in 1990 by value (£ million)

Alcoholic beverages (inc. duty)*	21 864
1 Meat and meat products	9 882
2 Dairy products (inc. ice cream)	5 884
3 Soft drinks	5 286
4 Fresh fruit and vegetables	4 598
5 Bakery products	3 790
6 Confectionery	3 700
7 Frozen foods	2 250
8 Canned foods	1 686
9 Fish and fish products	1 606
10 Snack foods	1 287
11 Hot beverages	1 281
12 Cereal products	1 230
13 Oils and fats products	971
14 Ready meals	715
15 Meal accompaniments	678
16 Sweeteners, preserves	535

(b) Increase of market sector values in the UK from 1989 to 1990 (per cent)

Alcoholic beverages*	8.2
1 Frozen foods	11.2
2 Soft drinks	10.9
3 Ready meals	10.5
4 Confectionery	8.8
5 Fresh fruit and vegetables	8.2
6 Snack foods	6.5
7 Cereal products	5.4
8 Meal accompaniments	4.3
9 Hot beverages	4.0
10 Dairy products	3.5
11 Meat and meat products	3.5
12 Fish and fish products	3.3
13 Bakery products	2.1
14 Oils and fats products	1.4
15 Canned foods	1.1
16 Sweeteners, preserves	0.6

Notes:

* As much of the market value of the alcoholic beverages sector is the duty, the value has not been used to rank this sector.

1. For the cost/benefit assessment for a potential instrument development, the market values listed here need to be seen together with other factors such as the relative values of raw materials and final product or the growth rate and profit margins in a sector. For snack foods, for example, the added value would, in general, be higher than for canned foods or meat. Additionally, the viability of an instrument development is increased when it is relevant to health and safety (where legislation is linked to the availability of instruments) or to the price of commodities (for example, in the case of the water content of wheat) or where the specifications for the proportional content of an expensive ingredient need to be met.
2. All market sectors increased in value, but not all growth rates exceeded the rate of inflation.
3. Categories are not mutually exclusive.

microelectronics have provided fast data processing and have made efficient process control systems possible. In the 1980s, programmable logic controllers (PLCs) were widely installed in the food industry. Massive control centres were designed earlier for integral plant control; these centres were subsequently replaced by distributed control systems (McFarlane 1983; Vidal 1988).

Whichever control system is used, it still has to make do with a small number of continuously updated product variables, and often relies largely on inputs at long time intervals and with long delays depending on the assay time and the distance to the quality control (QC) laboratory. The effective application of both established and novel sensors and instruments will play a key role in gaining the full benefit of the potential that modern control systems offer.

Table 1.1 shows the sizes and growth rates of food and drink market sectors in the UK.

1.1.2 On-line, at-line and off-line instrumentation

For optimum quality assurance the manufacturer requires cost-effective methods for the rapid assessment, and preferably the on-line measurement, of the chemical and physical properties and the microbial status of raw materials, process streams and end products. Monitoring during the processing operation helps prevent expensive rework or disposal of out-of-specification product. Tight control is needed for variables that influence the stability of the end product towards microbial spoilage or oxidative rancidity. This concerns particularly the monitoring of temperature profiles during heat processing and storage, the control of cleaning-in-place procedures, and the measurement of the pH, water activity, solute concentration and preservative levels of the product. Water activity, usually measured as equilibrium relative humidity (ERH), cannot be measured rapidly. From an on-line measurement of the moisture content, the ERH can be deduced if the isotherm is well defined.

The trend towards continuous automated production in place of batch processing necessitates tight feedback loops based on on-line monitoring methods or, failing that, on rapid at-line techniques. Even when a laboratory method provides a result within one hour of taking a sample from the line, over a tonne of product or over 10 000 jars, tins or packs of food may already have passed the production line. The cost of rework or disposal for such a quantity is considerable. Alternatively, excessive safety margins with respect to legal requirements or customer specifications on the minimum content of expensive ingredients will lead to an uncompetitively priced product.

Prolonged holding times to await the outcome of assays, as a regular part of the process, lessen the benefits of continuous processing. Nevertheless, holding times of around eight hours are currently observed prior to filling certain sterilized foods, for example, in order to await test results from impedance monitoring for microbial assessment (Chapter 17). Refinements of this technique, based on more sensitive oscillometric detection of impedance changes with microbial growth, for example, have been investigated in order to shorten the assay time (Cossar *et al.* 1990).

The advances in plant for automated continuous production and in the signal processing capabilities of process control systems have stimulated progress in the development of many novel sensors and instruments for the food industry, often by technology transfer from other industrial sectors or from the clinical sector (Kress-Rogers 1985, 1986). These have since matured; sensor concepts have been developed into prototypes, and instrument types already available in the 1980s have become more

versatile and can now be applied reliably to a wider range of foods and processing situations or determine a wider range of target variables.

With the help of these advances in on-line and at-line instrumentation (Fig. 1.1), quality assurance (QA) is employed increasingly in the management of manufacturing operations. The quality control (QC) laboratory supports QA by checking and updating the calibration of on-line and at-line instrumentation and by providing a wide range of analyses and assessments that are not feasible for QA implementation.

The variables measured on-line and those measured off-line in the QC laboratory do not necessarily coincide. The process stream at the on-line measurement point will often be quite different from the sample taken to the laboratory, either due to changes during sampling and transporting, or because the laboratory test measures properties of the end product, whereas the on-line instrument measures precursors of these, or other properties of the process stream or the process environment that will determine the relevant properties of the product.

When the time taken for a QC laboratory result exceeds a day, as would be the case for many microbiological tests or trace analysis assays for toxins, it is often impractical to hold the food in quarantine during this time, as a perishable food may be well into its shelf-life by the time the result is available. Even when prolonged holding times can be observed, it is not usually possible to provide 100 per cent screening of the product with QC methods, and so a negative result is no absolute guarantee that the whole production volume is 'clean'. The test then becomes a means of checking that good manufacturing practice (GMP) is being observed, and the process has to be analysed to define the product and process variables that can be monitored and controlled in order to minimize the possibility of manufacturing a product having too high a microbial load, carrying pathogens or containing toxins. This approach is known as hazard analysis critical control point (HACCP) system.

For the overall control of the process, the monitoring of level and flow rate as well as pressure and temperature are essential (Chapters 10–13). Important for the stability of foods towards microbial spoilage are product properties such as the water activity and the pH as well as the microbial load and the concentration of preservatives and nutrients (Chapters 16, 17, 20–23). The integrity of the food packaging is also vital, and in modern modified atmosphere packs (MAPs) the initial headspace gas composition and its retention during distribution and storage will be relevant. The adherence to appropriate storage temperatures (and ambient humidities) throughout the shelf-life needs to be ensured. An important process variable influencing the shelf-life is the time-temperature profile of the process stream and, related to this, the excess pressure in the headspace. Also relevant are the concentration and temperature of cleaning liquids and their efficient application to process plant surfaces (Chapters 10, 12, 13).

In conventional cooking and canning operations, heating the interior of a solid food item (or a highly viscous liquid) relies on thermal conduction, often resulting in overcooking of the outer layers in order to ensure adequate temperatures in the centre. This is not to say that high surface temperatures are not desirable in processes such as roasting, where the Maillard reaction provides a range of flavours and colours in the presence of reducing sugars and amino acids at elevated temperatures. The flavour changes caused by prolonged boiling are, however, usually considered undesirable. Microwave or radiofrequency waves, on the other hand, can penetrate food and heat deeper layers directly. Direct ohmic heating is also possible by mounting electrodes in contact with a conductive food, and ohmic heaters allowing continuous automated heat processing are available.

With these methods, it is possible to retain more of the flavour and vitamins of the food, and yet to ensure a given minimum temperature to be reached throughout. In order to optimize such processes for the manufacture of products that combine adequate cooking, pasteurization or sterilization (as required for the product) with good flavour retention, analysis of the spatial distribution of the time-temperature profiles is necessary.

Several variants of time-temperature integration are used to assess the effect of heat processing on a food. The most common is the F_0 value, which expresses the degree of sterilization of a food. The F_0 value (expressed in minutes) is obtained by calculating the integral

$$F_0 = \int L \, \mathrm{d}t$$

where $\lg L = (T - T_{\mathrm{ref}})/Z$ defines the lethality L, and the temperature T has been measured in the coldest part of the food. For F_0 evaluation, $T_{\mathrm{ref}} = 121^{\circ}\mathrm{C}$ and $Z = 10$.

For canned foods and ultra heat treated (UHT) products, F_0 values of 3 to 18 are used, depending on the types and numbers of spores present. This treatment results in commercial sterility, that is the remaining microorganisms will not cause spoilage or disease or have a detrimental effect on the product quality during its stated shelf-life (usually in excess of six months) (Lewis 1987).

Other values of T_{ref} and Z apply for the loss of nutrients by protein denaturation, vitamin destruction and certain other chemical reactions. A cook value can be defined, in analogy to the sterilization value F_0, to quantify the degree of cooking or overcooking and thus predict the loss of quality (flavour, nutrient levels) by heat processing.

Figure 1.2 summarizes the roles of on-line, at-line and off-line instrumentation in process management, quality assurance and quality control. Measurements relevant for product safety, stability and quality and for process management are listed in Tables 1.2 and 1.3. Instrument requirements and measurements for special concepts are given in Tables 1.4 and 1.5.

1.1.3 Technology transfer: opportunities and pitfalls

Instruments for measurements in quality control and in the control of processing operations in the food industry are often the result of technology transfer from other industries. The history of such new introductions has, in some cases, been characterized by initial successes, followed by a phase of disappointment with the instrument performance when the range of applications was widened. Subsequently, lost confidence had to be regained by defining the range of suitable application areas and by adapting the instrument or the setting-up and running procedures to particular applications. To avoid setbacks, it is necessary to understand both the instrument design and its underlying principles as well as the properties of the food and its processing environment.

Problems have, for instance, been experienced with some early applications of ultrasound flow meters in the food industry. These flow meters have the attraction of providing a non-contact measurement which facilitates maintenance and is intrinsically hygienic. However, for certain food process streams, unacceptable errors in the readings were observed until it was recognized that special designs or other types of flow meters were needed for samples with non-Newtonian flow profiles, or containing large particulates with flow rates differing from that of the carrier liquid, or where high attenuation of the ultrasound signal by the food liquid restricted the sampled flow volume to the outer layer (Chapters 12, 13, 15).

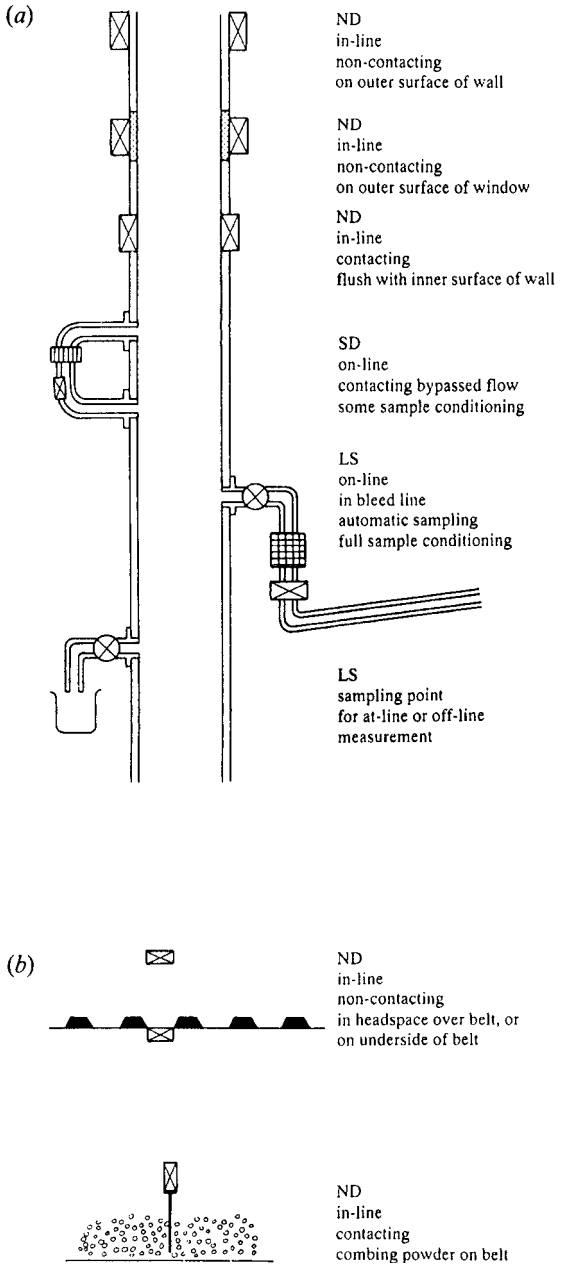
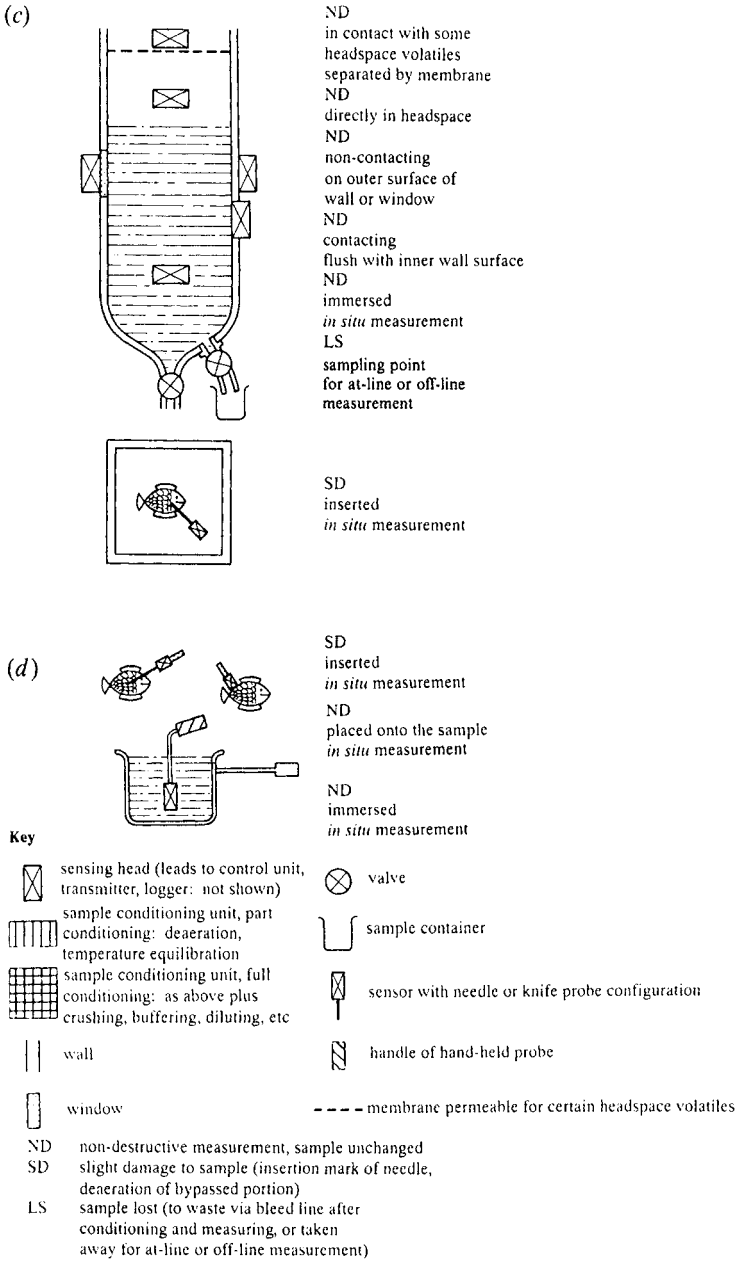


Fig. 1.1 Sensor configurations: (a) sensors on continuous processing lines (b) sensors on conveyor belts (c) sensors in batch processes (d) handheld sensors. The window material will depend on the instrument principle, for example Teflon for microwave transmission. Conditions for at-line measurements (not on-line, but in the production area) are more stringent than for off-line measurements (in the QC laboratory). At-line instrumentation and accessories should be free of glass components (potential foreign body hazard) and of toxic reagents that are not fully contained at all times. Also, mechanical robustness, tolerance of the processing environment (for example, of steam) and simple and rapid operation are essential.



1.2 Challenging conditions for sensors

1.2.1 Complex and variable samples

Many foods are highly complex in their chemical composition and in their physical structure. Gaseous, liquid and solid phases often coexist in the same product. Each phase may incorporate many different chemical compounds. One phase can be finely dispersed in another, or samples can be highly inhomogeneous or even largely separated. Within the

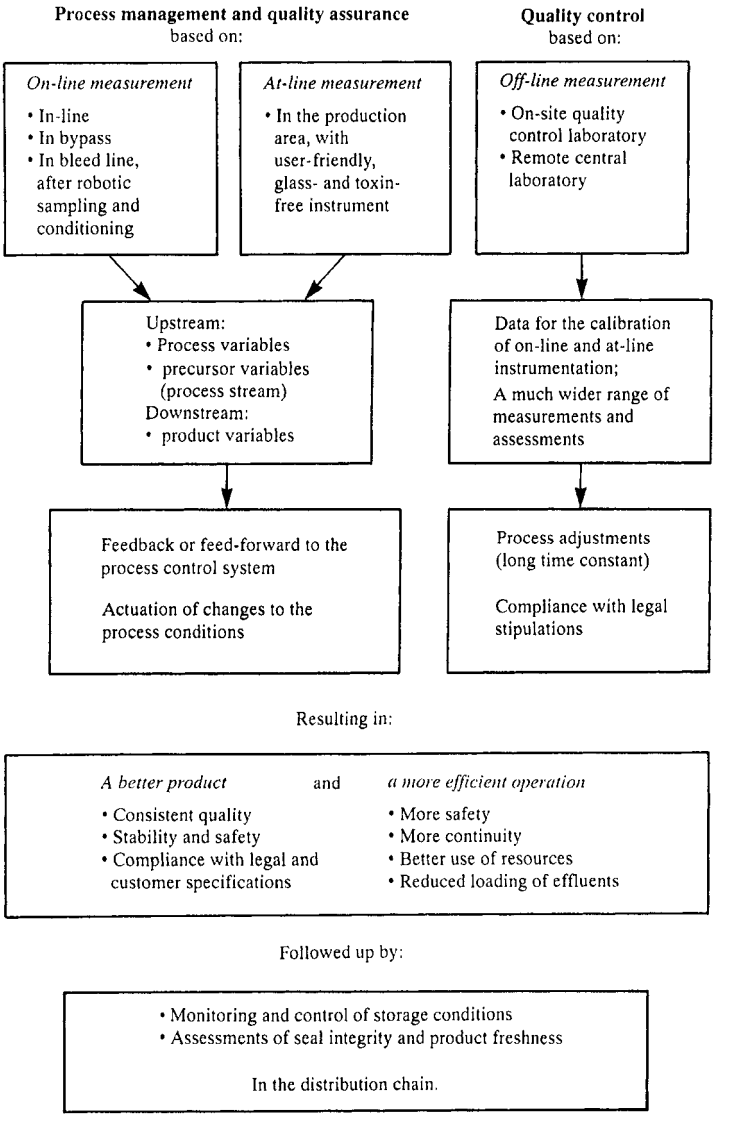


Fig. 1.2 Quality assurance (QA) and quality control (QC) in food processing operations.

Actuation of process changes

A programmable logic controller (PLC) may be used to operate actuators that cause changes in process conditions in dependence of a measured variable. Where complex relationships exist between the measured variables and the process, an expert system can provide an automatic evaluation of a set of measured values and a decision on corrections to the process conditions. Combined inspection/sorting systems are used to identify and remove, for example, products that contain foreign bodies or that are mis-shaped.

In-situ measurements in batch processes

The scheme needs to be adapted for industrial batch processing operations, for batch processes in catering establishments and for measurements in food distribution. *In-situ* measurements with dip- or stab-probes, or with instruments installed during the batch process or permanently in a processing vessel or storage container can be used here.

Calibration

The choice of the reference method can influence the calibration. Systematically different values can be obtained, for example, between drying and titration methods for moisture determination in the laboratory.

Table 1.2 Measurements in quality assurance and quality control

Measuring properties relevant for product quality

- appearance (colour, gloss, shape)
- texture, mouthfeel, pouring characteristics
- flavour (aroma, taste)
- nutritional value
- functional properties
- composition according to specifications

Screening for product safety
chemical contamination (agricultural residues, endogenic toxins, …)
microbial contamination (total load, presence of pathogens and spoilage organisms, …)
contamination with unwanted genetically modified organisms
foreign matter (metal or glass fragments, insects, stones, …)
unwanted matter (nutshells, fruit calices, …)

Assessing product stability towards
- chemical reactions (such as oxidative rancidity)
- microbial growth (due to inappropriate pH, water activity, preservative concentration, either in the product as a whole or in a small region within the product)
- microbial or chemical contamination (due to defective or inappropriate packaging) (including the migration of compounds in the packaging material into the food)
- migration of water or fat (between pastry shell and filling, between food and environment)
- loss of protective atmosphere (due to defective seal) (for products packed under a modified atmosphere designed to suppress microbial growth or oxidation)

liquid portion, fat and water may be combined in an emulsion, or even in a double emulsion. Water can be present as free water or bound in many different ways: as water of crystallization, bound to protein or starch molecules, entrapped in biopolymer networks or absorbed on solid surfaces of porous food powder particles. Active enzymes may be present, either in the tissues of fresh meat or produce, or within the cells of the microbial flora.

Table 1.3 Measurements in process management

Objectives
- ensure safety and continuity of the processing operation
- maintain conditions for in-spec. products
- use resources efficiently (labour, raw materials, energy, machinery)
- reduce loading of effluents (e.g. of waste water with organic matter)

Measurements
- pressure
- temperature (also spatial distribution of temperature and time integral over temperature),
- pH
- mass and volume flow rates of liquids and particulate solids
- fill levels of liquids and particulate solids bulk density, weight apparent viscosity

Also wanted, but more difficult to achieve on-line:
- chemical composition (gross and fine)
- complex rheological properties (yield value, elasticity, …)
- particle, droplet, bubble size, (average size and distribution)
- volatiles evolved in cooking, baking, roasting, drying operations

Table 1.4 Instrumentation requirements, on-line

- hygienic sensing head
- contaminant-free (no reagents, no microbes)
- no foreign body hazard (no fragile glass components), robust
- CIP (cleaning-in-place) tolerant if permanently installed on-line (alternatives for specific chemical measurements: instruments with disposable sensing element which must be easily replaced and inexpensive, or, in certain applications, robust, easily cleanable dip-probes or stab-probes for *in situ* measurements)
- reproducibility in accordance with task, reliable, low maintenance effort suitable for complex chemical and physical sample properties
- total cost (capital, maintenance, running) in good proportion to benefits

Samples in the food industry are, moreover, very diverse and highly variable. The season, the region of origin, the harvesting and storage conditions as well as the processing steps (such as the fermentation of cocoa beans) will all influence the properties of the raw materials. New food-processing technologies are being introduced to provide an ever wider range of food products that require frequent adaptation to changing consumer preferences and market structures.

1.2.2 Hostile conditions and stringent hygiene requirements

The pH extends over a wide range, with low values for vinegar or citrus fruit juices and high values for caustic cleaning solutions used regularly in-line. A wide range of pH values is also encountered in the monitoring of effluents, that is waste liquids formed in washing raw materials or in cleaning container surfaces, for example.

Table 1.5 Measurements for special concepts

HACCP – Hazard analysis critical control points
On-line measurements
- pressure, temperature (spatially resolved, time integral)
- relative humidity
- product pH product solute content
- strength and surface coverage of solutions used for periodic cleaning of machinery

Off-line measurements
- water activity (as equilibrium relative humidity)
- pH (spatially resolved) preservative concentration
- microbial contamination of ingredients including water
- microbial contamination on machinery and on other surfaces in the production area

Marker (indicator) approach
For the on-line, at-line or *in situ* assessment of
- microbial pre-spoilage status
- oxidative rancidity status
- level of heat-induced deterioration
- progress of ripening or conditioning browning potential
- end of heat processing operation

Measured are chemical or physical variables that have first been identified as indicative of the complex condition of interest. Usually, a given marker (or indicator) variable will be valid for a particular group of products only. (See Section 1.3.4 and Chapter 19.)

Temperatures vary from freeze-drying conditions (−50ºC or lower) to hot frying fat conditions (up to 250ºC) and roasting operations (320ºC or higher). Processing and packaging under vacuum is employed, and excess pressure is used in cooking and canning operations. A retort would typically operate with pressures of 60–600 kPa, that is 0.6–6 bar (McFarlane 1983, see Appendix B, Tables 3 and 4). Particularly severe conditions can prevail in a cooker extruder, where both high pressures (over 10 MPa, that is, over 100 bar) and high temperatures (around 200ºC) can be encountered. Moreover, the inner barrel surface is scraped by the extruder screw, and access to the food mix within the barrel or in the extruder head is certainly restricted. The food mix itself can be quite abrasive in the early part of its passage through the extrusion cooker. Maize grits, for example, may be present, expanding later on in the fashion of popcorn. These conditions present a challenge even for the design of pressure and temperature (p/T) probes. (For a description of extrusion cookers see McFarlane 1983; O'Connor 1987; Wiedmann and Strecker 1988.)

Nevertheless, sensors for the measurement of moisture and other variables are under development for this hostile environment. Radiofrequency open-ended coaxial probes have been designed to fit into the openings foreseen for the bolt-type p/T probes designed for extrusion cookers, and a microwave stripline has been constructed for mounting in an extrusion head (Chapter 9).

In general, the conditions in the food industry are more favourable than in a cooker extruder. A common challenge for *in situ* sensors is, however, the cleaning-in-place (CIP) procedures used in many processing systems in the food industry (Kessler and Weichner 1989). These usually involve flushing with hot caustic soda solutions (NaOH) which can corrode probe surfaces, and this is particularly unfavourable for many chemical sensors. High-pressure steam cleaning is another effective CIP procedure; this will challenge the mechanical and thermal stability of a sensing head.

The strict hygiene standards in the food industry also demand that in-line probes in contact with the sample must have crevice-free surfaces. This applies both to the sensing head and to the mounting flange area. For aseptic processes, any sensor surfaces in contact with the sample need to be tolerant to CIP procedures. In fermentation processes, the use of a disposable sterilized sensor can be an option.

Any danger of chemical contamination of the food by sensor reagents or components of slight solubility must be eliminated. The introduction of foreign bodies, particularly glass or metal fragments, in the case of damage to the sensor, must also be prevented. Food powders with a very low moisture content can accumulate high electrostatic charges, and sensors that may come into contact with such powders (typically starch-based products) must be designed to minimize the risk of a dust explosion.

The transducer and electronics may have to withstand exposure to water, steam or airborne dust. Occasionally, they may be enrobed in chocolate or coated with a thin film of condensed polymerized frying oil. Sensors in contact with food or food volatiles are often subject to fouling by proteins, fats or starch particles (Kessler and Weichner 1989).

Electromagnetic interference (EMI) will be encountered in industrial microwave ovens or in direct ohmic heating appliances. There will also be electromagnetic noise and mechanical vibrations from pumps, hoppers and other plant. Rotating mixer paddles scraping the walls of a vessel may be in the way of a radiated signal or restrict the positioning of a wall-mounted probe.

1.2.3 Non-contact techniques and robotic sampling and conditioning

Given the often hostile conditions for invasive sensors, either during the processing of foods or during the periodic cleaning operations, and the always stringent hygiene and other food safety requirements, non-contact measurements are particularly attractive to the food industry. These can be based on the interaction of electromagnetic waves, including gamma-rays, light, infrared radiation, microwaves or radiofrequency waves, or of ultrasound signals with the sample. Such methods do, however, require an awareness of the nature of the interaction of the applied signal with the food, its headspace and container. This understanding is needed at all stages of instrument development, in the choice of suitable applications and installation points, during the setting-up and calibration procedures (including the preparation of training samples), in the running of the instrument and in the evaluation of the readings (Chapters 6, 7, 9, 11, 12, 13).

To develop the wide range of sensors desirable for in-line measurement in the food industry would be prohibitively expensive. Not only different target variables, but also different analytical ranges and variable chemical and physical environments, would have to be catered for. The recognition of the cost that would be associated with the development of in-line sensors for a wide range of chemical, physical and microbial properties, each for a wide range of diverse applications, has led to an interest in techniques that make the best of the sensors available.

Robotic sampling and sample preparation systems allow rapid measurements at short intervals by enabling the use of sensors that would otherwise be confined to laboratory applications. This approach has been implemented particularly in Japan. An example is shown in Fig. 1.3 (see also Chapter 20, Section 5).

1.3 Interpreting the readings

1.3.1 Measured variables and target variables

In non-contact measurements, the relationship between the measured variables and the target variables is often complex, so that a given calibration will apply only to a limited range of food products and processing conditions. For instance, a water content measurement based on near infrared reflection analysis will have to rely on a predictable relationship between surface moisture and average bulk moisture content; or, the monitoring of solute concentration by a measurement of ultrasound velocity depends on a

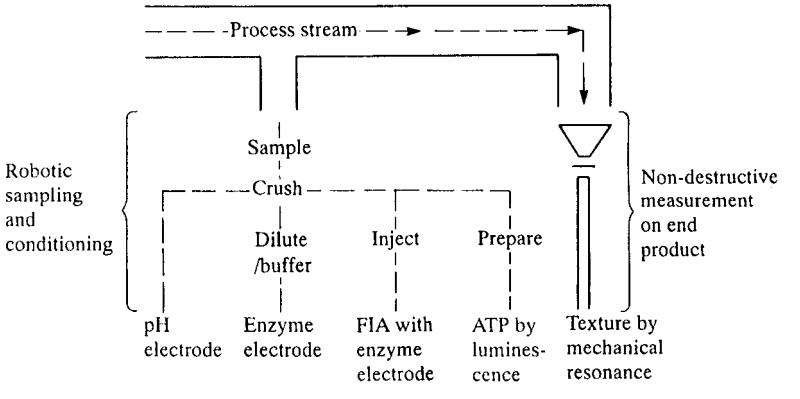


Fig. 1.3 Robotics approach.

constant composition of both the solute and the carrier liquid (and on compensation for temperature changes, as is the case with most measurement methods). Care in the setting up and calibration are essential, as is the choice of appropriate applications. Non-contact in-line techniques will then provide highly reliable continuous measurements that allow process adjustments before an out-of-specification situation arises (Chapters 6, 9, 12, 13).

In contact measurements also, the measured variable is not always the target variable. For example, pH is often measured as an indicator of acid concentration (provided that the acid composition is known). Ion activity is often measured in place of ion concentration. Frying oil samples are taken for an at-line measurement of colour or free fatty acid (FFA) content in order to infer the degree of frying-induced polymerization and oxidation. Yet, both colour and FFA-content are highly dependent on other factors such as the oil type, the food fried and the frying conditions. (See Chapters 19 and 20 on oil quality and pH, respectively.)

In the QC laboratory, the assay of chemical composition can involve deductions from the proportion (by weight) of sample becoming volatile or dissolved under certain conditions. Clearly, such assay types, and many others, need to take into account the nature of the sample. Indeed, the official methods prescribe sample-specific assay procedures.

In addition to these relationships between measured and target variables, the significance of a target variable for the manufacturing operation needs to be considered. This can reside in assuring the safety and stability of the food, the enjoyment in handling and eating it, and the efficiency in producing it or in complying with legal regulations or customer specifications. These aspects are discussed in Section 1.4.1.

1.3.2 Relationship between in-line and QC laboratory methods

Differences between the readings of the in-line instrument and the off-line quality control (QC) laboratory results are, at times, unjustly blamed on the in-line method. The QC reference assay can be applied only to a small fraction of the sample volume passing the sampling point, and this alone can lead to a result differing from an in-line method that provides 100 per cent screening. Moreover, different laboratory methods will often give systematically differing results between them for a particular food sample type, even though they may give identical results for other sample types (see, for example, Fig. 1.4).

Often, the definition of the measured variable depends on the laboratory reference method used. For example, for an oven-drying procedure, moisture is defined as that part of the sample that will be driven off at the applied temperature and pressure. For a titration procedure, on the other hand, the relevant part of the water is that which can be extracted from the food matrix (or dissolved and dispersed together with food solids) and brought into contact with the reagent. For this reason, official analysis protocols exist for different food types; for accurate results to be achieved, an instrument used in the QC laboratory needs to be calibrated for each food type against the official method.

In some situations, the comparison between the on-line reading and the chemical or rheological QC result, for example, is not strictly valid because the sample changes on removal from the line and during the preparation and analysis steps in the QC laboratory. Oxidation, thinning, thickening, fermentation or other changes can occur during this time, and homogenization steps can cause the breaking up of tissue cells, leading to changes in the composition of the juices and also exposing the cell contents to oxidation or other reactions.

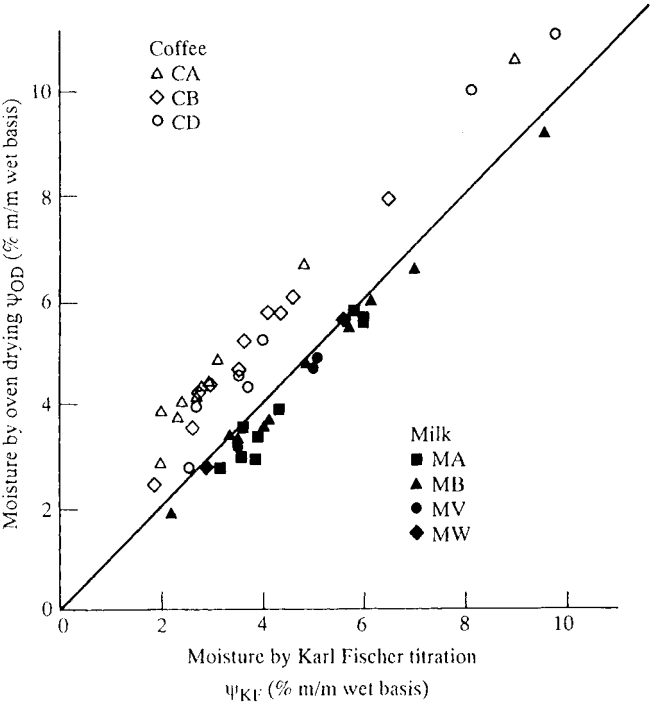


Fig. 1.4 Comparison of laboratory methods for moisture measurement. Oven drying at 102ºC to constant weight leads to an increased apparent moisture content when non-water volatiles are present. This can be avoided by vacuum drying at a lower temperature. When lactose is present, incomplete dehydration can decrease the apparent moisture content by oven drying. This is avoidable by moistening prior to oven drying. Symbols mark different brands of instant coffee and milk powder (Kress-Rogers and Kent 1986).

When changing from a process control system, updated from time to time by data from the QC laboratory, to a system relying primarily on continuous feedback from in-line instrumentation, the significance of the measured properties for the food processing operation needs to be well defined. It may well be favourable to replace an off-line measurement for a given set of variables with an on-line method determining a different set of variables, at a point upstream from the existing QC sampling point. Feedforward control based on the continuous on-line monitoring of the process stream before and during processing can often provide tighter control of the end product properties than delayed feedback control based on an off-line measurement of the end properties themselves.

Based on an analysis of the process as a whole, together with an awareness of the options for on-line and at-line instrumentation, an effective data input system for process control purposes can be designed. A complementary range of laboratory instruments will provide calibration updates and a wider range of measurements (see the last entry in Appendix A).

1.3.3 Data processing approaches

When several on-line measurements are carried out simultaneously, there is not necessarily a one-to-one relationship between each measured variable (or a combination of these) and a corresponding control parameter to be adjusted. The design of the

response pattern of the control system can then be based on a variety of data processing approaches. Ideally, but rarely, there is a mathematical model based on sound physico-chemical principles, which links the measured variables to the relevant properties of the end product and to the process parameters that can be adjusted. For other cases, data interpretation based on neural networks (Gardner and Hines 1997, Section 5; Jansson 1991) and control decisions based on fuzzy logic (Berrie 1997; Eerikainen *et al.* 1988) as well as neuro-fuzzy systems (Theisen *et al*. 1998; Tilli 1996) have become established. (See also the entries on fuzzy logic and neural networks in Appendix A to this book.)

Neural networks in particular have attracted much attention in recent years. Here, several on-line sensors that are not specific for an intended property are combined. Extensive output data from these sensors and associated quality control results are then collected over a period of weeks or months. These data are subjected to analysis with neural networks. This can result in a signal processing mode learnt by the neural network. The output of the educated network will then be indicative of a characteristic of the finished product such as taste or aroma, although none of the contributing sensors necessarily has a well-defined physico-chemical relationship to the characteristic of interest.

There is a temptation to employ such systems as a replacement for the development of sensors and instruments that give an output with a well-defined relationship to the process and product. However, the short-term saving in avoiding sensor development can be outweighed by the cost of frequent updating of the signal evaluation procedure for non-specific measurements. This updating procedure may involve the collection of process measurement data and associated product assessments by QC laboratory tests and taste panels over many weeks, and may be required whenever raw materials or recipes change. When the need for an update is not recognized in time, the system will be unreliable, and the quality of the product will be adversely affected. Although sophisticated data evaluation systems will play an important role in the future, they will be complementary to rather than a replacement for measurement techniques based on sound physico-chemical principles. (See also Chapter 20, Section 20.5.2.)

A special case is the assessment of aroma and taste. Odours and aromas are usually composed of hundreds of compounds carried in an air stream made up of further compounds. Additive, synergistic, antigonistic and compensative effects can occur between the chemical components contributing to a flavour, that is the aroma and taste as perceived by the human nose and tongue. A sensor that is highly selective for just one chemical compound is not meaningful in the assessment of flavour.

The standard approach to flavour analysis in the food industry is the organoleptic panel composed of six or more carefully screened and well-trained panellists and following elaborate procedures (see Chapter 2). Due to the cost and time required for this approach, it is suitable for periodic quality control but not for continuous monitoring or field work. To address this problem, electronic noses and tongues have been developed. These are modelled on the mammalian nose and tongue but of lesser complexity, and each of these systems is intended for a specific range of applications. They are based on arrays of (typically about 20) sensors with broadly overlapping specificities combined with pattern recognition methods including neural networks (Chapter 19; Kress-Rogers 1997).

1.3.4 The marker approach; novel sensors

In the case of lengthy tests for properties such as freshness, the progress of microbial or oxidative degradation processes, or taste and texture, efforts are increasingly made to

identify and measure chemical or physical indicator variables linked to the condition. This can be described as the marker approach. For example, prototype probes for meat freshness and frying oil quality assessment have been developed, based on a biosensor array and a vibrating rod viscometer, respectively. The assessment of meat and fish freshness by indicator gases and volatiles or by compounds involved in the decomposition of ATP have received increasing attention in recent years (Chapters 19, 20, 22).

Novel chemical sensors, including biosensors (Turner *et al*. 1987; Scheller and Schmid 1992; Kress-Rogers 1997), have been developed in the last four decades, and the pace of this development has gained momentum. A sizeable number of biosensors have already been adapted for food applications. Many other sophisticated biosensor adaptations for the food industry are still at the prototype stage. This is consistent with the delay between the research and application stages of clinical biosensors, for example, which have received attention and funding earlier and more lavishly. In the absence of fully optimized biosensors for food applications, many current practical biosensor applications in the food industry employ the robotic approach (see above).

Earlier, the microelectronic pH and ion sensors (primarily ion-selective FETs or ISFETs) had a hesitant commercial development. In recent years, however, commercial instruments designed for the food industry have been available and have solved many of the problems experienced with traditional pH and ion probes (hazard of glass membrane, laborious maintenance, etc.). Fibre-optic probes for chemical sensing are continuing to make progress. They are immune to EMI (electromagnetic interference) in microwave ovens, and can be configured as robust and mechanically flexible remote probes. Optical sensors are also the basis of novel immunosensors (Chapter 20).

1.3.5 Instrumentation as an interdisciplinary subject

The development and implementation of successful new measurement applications are tasks requiring effective interdisciplinary communication and cooperation. This can be accomplished only when instrument suppliers and users have a common basis of understanding of the interplay between the physical, chemical and microbial properties of the food and its environment on the one hand, and the physics, chemistry and electronics governing the instrument characteristics on the other. An awareness of the nature of the reference method and of the significance of the target variable for the food processing operation is also needed.

In moisture measurement, for example, one has to establish whether the total water content is most relevant, or whether only water molecules with a high degree of mobility play a role. This will depend on whether the measurement is undertaken on account of legal requirements regarding the maximum water content of, e.g. margarine, for the price definition of commodities such as wheat, or with a view to the influence of the water content on the product texture or its relationship with the water activity. The last-mentioned factor determines the stability towards microbial spoilage and water migration from pie filling to pie shell, for example. Water activity is usually determined by a measurement of the equilibrium relative humidity of the sample, which by definition cannot be measured instantaneously. Instead, the water content can be measured in-line, and the water activity can then be deduced for a given product with a given processing history. Moisture and microbes are both ubiquitous, and improved instrumentation for their determination has been a prime concern for the food industry for many years (Chapters 6, 9, 12, 13, 16 on moisture and water activity, Chapters 17, 19–23 on specific microbes and microbial load).

From the instrument engineer's point of view, it may seem surprising, at first glance, to find methods based on a physical interaction of the sample with the instrument (using visible, near infrared, microwave or ultrasound waves, or employing mechanical stress; Chapters 3–9, 12–15, 18) and those based on chemical interactions (Chapters 19–23) presented together. The food scientist, on the other hand, may prefer to consider instruments for the determination of physical properties (rheology, colour, density, particle size) of the sample separately from those for the assay of chemical or microbiological properties. Process control engineers may wish to separate on-line and at-line instrumentation from off-line instruments. However, in formulating a system of quality assurance and quality control procedures for a food manufacturing process, it is advantageous to be aware of the alternative techniques, each with its own characteristics, for the determination of a particular sample property (for example, for sugars, see Chapters 6, 8, 12, 13, 20–23; for water content see Chapters 6, 9, 12, 13). A number of instrument types allow the measurement of both physical and chemical properties. For example, bulk density and water content can be determined simultaneously by a microwave technique (Chapter 9; Kress-Rogers and Kent 1987); film thickness and composition can be measured with near infrared instruments (Chapter 6).

Also, there is often a choice between process control based on measuring a certain set of variables upstream and in-line (for example, pressure, temperature, pH), and that based on measuring another set downstream and off-line (for example, glucose/sucrose ratio). As a further option, some off-line instruments can be used at-line with the help of robotic sampling and sample preparation, thus providing rapid feedback based on a wider range of measured variables. On-line and QC laboratory methods have to interlink in guiding process control, and are therefore presented together in this book, with the emphasis on the interplay between the instrumental method, the food and the manufacturing process.

1.4 Measurement types

1.4.1 Target variables

The on-line or rapid at-line measurement of variables relevant to the eating quality, the wholesomeness and the safety of the food product is an increasing concern of the food industry, and this is reflected in the topics covered in this book. Colour and other aspects of the appearance are dominant in determining the first impression and influence the choice of food products by the consumer. Aroma, taste, and texture or mouthfeel influence the enjoyment of the food and determine whether the consumer comes back for more of the product. The rheological properties also affect the handling characteristics of the product both in processing and in the hands of the consumer.

All these food properties can be assessed by panels of trained persons, or animals in the case of petfood (Chapter 2). However, for frequent control purposes this is not practical, and instruments for the measurement of colour attributes, of variables related to the chemical composition (either proximate analysis or more specific and sensitive measurements) and of those related to the physical structure of the food (rheology, particle size, new methods in texture assessment) are used if equipment of the desired specifications is available (Chapters 3–5, 6, 8, 14, 15, 18, 20–23). Electronic noses and tongues have been developed in recent years to mimic the human (or pet) sense of smell and taste for specific applications (Chapter 19).

The assessment of wholesomeness and safety is not generally possible or advisable with the human senses, and instruments or laboratory tests will always be required for

Table 1.6 The significance of measurements for the manufacturing operation

	Food safety and stability			Food quality					Production efficiency		Compliance		
	Microbial aspects	Chemical aspects	Physical aspects	Nutritional aspects	Appearance	Texture consistency	Aroma	Taste	Ingredients management	Process management	Legal conditions	Customer specifications	Labelling regulations
Colour					×							×	
Sorting by colour	×	×	×	×	×	×	×	×			×	×	×
Temperature	×	×	×							×			
Temperature – time integral	×	×	×	×	×	×	×	×			×	×	1
Pressure	×	×	×							×			
Level and flow rate									×	×			
Particle, droplet or bubble size			×		×	×			2	×		×	
Solid/liquid ratio and crystal size			×		×	×				×		×	
Bulk density			×		×	×			×	×		×	×
Rheology			×		×	×			3	×		×	
Texture						×				×		×	
Water activity	×	×	×		4	4	4	4		×	×	×	1
Water content	5	5	5	×	×	×		×	×	×	×	×	×
Proximates: fat, protein, carbohydrate, ash				×	×	×	i	×	×	×	×	×	×
pH	×	×					i	i		×	×	×	1
Acidity				×			×	×				×	
Sodium, potassium, calcium				×				×				×	×
Specific sugars	6		×	×	×	×		×				×	

Total reducing sugars					7		7	7			×				
Solute concentration	6		6	×	×	×		×			×		×		
Alcohols				×				×				×	×	×	
Preservatives, antioxidants	×	×		8	8		8	8				×	×	×	
Vitamins, trace minerals				×				×				×	×	×	
Emulsifiers, gelling agents			×		×	×					×		×	×	
Flavourings, colourings				×	×		×						×	×	
Toxins, residues		×										×	×	×	
Volatiles during cooking							×				×				
Volatiles during storage	×	×		×			×					×	×		
Microbial load, contamination	×	×			×		×	×				×	×		
Biomass, functional	×	×	×	×	×	×	×	×	×	×			×	×	
Chilled meat freshness	×	×		×	×		×			×	×	×	×	1	
Frying oil quality		×		×	×	×	×	×	×	×	×	×	×	1	

Key

× direct link; i strong indirect link; 1 influencing storage conditions or use-by date; 2 influencing packing density; 3 influencing enrobing thickness; 4 governing microbial growth and metabolism as well as chemical reactions and water migration processes, thereby influencing composition and texture; 5 linked to water activity for a given food with a given process history; water content is more easily measured at-line and can be used as an indicator of water activity under controlled conditions; 6 solutes such as sugars or salts reduce water activity and have traditionally been used to preserve foods (for example jam, bacon); 7 at high temperatures, reducing sugars react with amino acids to form a wide range of compounds that flavour and colour roasted and baked products (Maillard reaction); 8 for example, antioxidants reduce the rancidity of oils and the browning of produce.

Authenticity assessment. The comparison of the sample's free fatty acid composition, ultraviolet absorption spectrum or other 'fingerprint'-type assays with the pattern expected for the claimed food type and origin is used in the authenticity assessment of foods.

these food properties, including the content of nutrients, microbial load, chemical contaminants, pathogenic microbes and foreign bodies. Not only the current status but also the stability of the product during the intended shelf-life needs to be assessed. (On the shelf-life of foods, see for example Chapter 7 in Stewart and Amerine 1982.) To this end, a determination of the product's water activity, pH, packaging integrity, existing flora, content of preservatives, antioxidants content and other factors is undertaken.

The temperature of a food needs to be monitored in heat processing (such as sterilization), tempering and conditioning and also in storage. Spatial resolution is often needed, particularly information on the hottest and coldest points of the sample. A time-temperature profile including integral, maximum and minimum is also an important measurement target. Remote infrared thermometry has the advantage of providing a non-contact measurement and is also the basis of thermal imaging systems (Chapter 7). A development for depth profiling is the distributed fibre-optic temperature probe. Table 1.6 shows the significance for the manufacturing operation of measurements of a range of variables.

1.4.2 Instrumental methods

On-line instruments based on the interaction of electromagnetic waves or ultrasound with the sample are discussed in the first part of the book. In the second part, instruments in the quality control laboratory are treated. In the third part, the expanding range of chemical sensors, including biosensors, is presented. The development of sensors for the identification of an indicator variable that is more amenable to on-line implementation than current chemical reference methods for quality assessment is discussed. The assessment of sensory properties by both sensory panels (composed of humans or pets) and by instruments (optical instruments, electronic noses and tongues, texture analysers) are detailed. Instruments that require an understanding of their interaction with the food and the manufacturing process as a condition for successful application are the subject of this book. In order to provide a useful description of principles and food-industry-specific application details for such instruments, it has been necessary to choose certain instrument types to the exclusion of others.

Only instrumental methods that can be understood fully on the basis of classical physics have been included in this book; thus instruments based on nuclear magnetic resonance (NMR) or electron spin resonance (ESR) are not represented. This is not to imply the lesser importance of these methods. In fact, the development of new applications of NMR and NMR imaging (NMRI) to the food industry has progressed rapidly, and provides the basis for many new instruments (see, for example, Levine and Slade 1991, pp. 405–626). For the recognition of irradiated foods, ESR applications are being developed.

Pulsed NMR techniques are already widely used in quality control in the food industry for the determination of water and lipids content and of the solid/liquid ratio of fat. Bench-top instruments for these applications are readily available and are being further optimized for a wider range of food applications (including the investigation of water 'binding' to biopolymers) and more convenient operation. Furthermore, research into the NMR characteristics of foods and the development of novel magnet designs for NMR instruments are now underway. This will lead to new application areas for NMR measurement in the food industry. One area is the on-line implementation of bulk NMR measurements as they are currently undertaken in QC laboratories. These are to become feasible for a much wider range of process lines than is possible at present. They will aid

in the control of many operations in food processing, including baking, drying, concentrating, freezing, thawing and tempering. The other area is the use of NMR imaging in product development and quality control tasks. This latter measurement type will, in the medium term, be restricted to central laboratories owing to the cost and complexity of the equipment.

Also omitted in this book are those instruments that rely on ionizing radiation (such as x-ray foreign body detection systems) or on a substantial computing capacity, as, for example, image acquisition and analysis systems. Real-time processing and automatic evaluation of visible light, x-ray, infrared or other images is another area that is expanding rapidly. The underlying principles and application details relevant to the food industry for these systems could best be presented in a book dedicated to this area alone.

The appendix to this chapter summarizes measurement types.

1.4.3 Fringe benefits

The dielectric properties of food, discussed here in the context of measurements based on microwaves, are also relevant in the application of such radiation at much higher intensities to the processing of foods in industrial and domestic microwave ovens (Ohlsson 1988). Indeed, many dielectric data for foods have been acquired in order to predict microwave heating characteristics. For food scientists, a familiarity with the principles underlying dielectric measurement affords the added benefit of helping in the analysis of processing characteristics. To a lesser extent this applies to ultrasound, which also has applications both as a low-intensity signal used in measurement and as a high-intensity irradiation employed in sonoprocessing and sonochemistry. The overlap between the food properties relevant for measurement and those relevant for processing is smaller here. Ultrasound processing is therefore briefly discussed as a separate topic (Section 13.2.7).

1.5 Further reading

The emphasis in this book is primarily on the measurement of food properties as this requires a particular understanding of the interplay between the instrumental method and the food and its environment. However, this understanding is also required in the measurement of certain process variables such as the volume and mass flow rates of food liquids with complex rheologies. Ultrasonic measurement of volume flow rate and of liquid and solid level is discussed in this book (Chapters 11–13).

Volume flow rates do not always provide adequate information for materials with uneven aeration or temperature distribution, and are also difficult to measure for non-Newtonian liquids (Chapters 14, 15) or inhomogeneous materials. For materials such as molten chocolate, mass flow rate can now be measured successfully with Coriolis force flow rate meters (developed by Exac and Micromotion). These have the form of a U-shaped tube (or two parallel U-tubes), which can be flanged into the pipeline without restricting the flow. The Coriolis technique has been reviewed by McKenzie (1990), with details on one of the two embodiments.

A straight vibrating tube section can be flanged into pipes for measuring the mass density of the process stream. The tube section is isolated vibrationally from the remainder of the pipe by bellows. Another sensor based on mechanical resonance is the vibrating rod probe for the measurement of level, density or viscosity. Such probes are

available in dipstick form or in flange mountings. This sensor family is represented in the book in the form of a prototype probe for the *in situ* determination of frying oil quality, based on a measurement of the viscosity of the hot oil by a vibrating tube viscometer (Chapter 19). A review of the principles underlying the function of a wide range of mechanical resonance sensors has been given by Langdon (1985).

For the measurement of temperature, both contact methods and a non-contact method that also lends itself to combination with linescan and imaging techniques is presented in this book (Chapter 7, 10). Fibre-optic thermometers and quartz temperature sensors (shear mode resonators or tuning fork devices), which have become available for industrial applications in recent years, are reviewed by Schaefer (1989). Updated overview tables on the specifications of conventional contact thermometers such as thermocouples are published regularly by transducer magazines. This is also the case for strain gauge pressure transducers. Information on special miniature designs for pressure and temperature transducers that can be mounted inside sample cans during heat processing can be obtained from the suppliers.

Humidity sensors have been discussed as parts of instruments for the measurement of water activity. Similar sensors can also be used in monitoring ambient humidity in many storage or processing areas. In the latter applications, the required accuracy is less stringent; however, the emphasis is on robustness and long-term stability with minimal maintenance. High temperatures often need to be tolerated, for example in oven flues (McFarlane 1983). Inexpensive miniature humidity sensors based on solid-state devices are now part of a number of domestic appliances (such as some makes of tumble dryers or microwave ovens).

An overview of the automatic control of food manufacturing processes is given by McFarlane (1983). He describes processes in the areas of raw materials handling, recipe dispensing, pre-processing, cooking processes, biochemical processes, finishing and packaging. The range and tolerance for the controlled variables are given for each process. There are also books giving details on processing operations in one particular sector such as fish canning (Wheaton and Lawson 1985). Some recent developments in the control and optimization of food processes are presented in symposium proceedings edited by Renard and Bimbenet (1988) and by Spiess and Schubert (1990).

A comprehensive compilation of instruments, each described briefly without industry-specific aspects or application details, is available in the form of a handbook on general instrumentation (Noltingk 1988). Often, instrument manufacturers will supply application details and related literature.

An introduction to food science and food technology has been given by Stewart and Amerine (1982). Information on the physical properties of foods can be found in the books by Lewis (1987), Jowitt *et al.* (1987), Singh and Medina (1988) and Okos (1986).

1.6 References

BAILEY, L., BOYLE, C. and HILLIAM, M. (1991) The UK food and drinks report: market, industry and new product trends. Leatherhead Food Research Association Special Report, April 1991.

BERRIE, P.G. (1997) Fuzzy Logic in the Evaluation of Sensor Data, pp. 469–500 in *Handbook of Biosensors and Electronic Noses: Medicine, Food and the Environment*, Kress-Rogers E. (ed.), Boca Raton, New York, London, Tokyo, CRC Press Inc.

COSSAR, J.D., BLAKE-COLEMEN, B.C., RAMSAY, C. and ATKINSON, T.A. (1990) Oscillometric instrument for the non-invasive detection of low-level microbial activity. *Biosensors and Bioelectronics*, **5**, 273–89.

EERIKAINEN, T., LINKO, S. and LINKO, P. (1988) The potential of fuzzy logic in optimization and control: fuzzy reasoning in extrusion cooker control, pp. 183–200. In *Automatic Control and Optimization of Food Processes*, Renard, M. and Bimbenet, J.J. (eds) Elsevier Applied Science, London.

GARDNER, J.W. and HINES, E.L. (1997) Pattern Analysis Techniques, pp. 633–52 in *Handbook of Biosensors and Electronic Noses: Medicine, Food and the Environment*, Kress-Rogers E. (ed.), Boca Raton, New York, London, Tokyo, CRC Press Inc.

GEORGALA, D.L. (1989) Modern food processing, in *Food Processing, Proceedings of the Ninth British Nutrition Foundation Branch Conference*, Cottrell, R.C. (ed.) Parthenon Publishing Group, Carnforth and New Jersey.

JANSSON, P.A. (1991) Neural networks: an overview. *Analytical Chemistry*, **63**, 357A–362A.

JOWITT, R., ESCHER, F., KENT, M., MCKENNA, B. and ROQUES, M. (eds) (1987) *Physical Properties of Foods Vol. 2*, COST 90bis Final Seminar Proceedings. Elsevier Applied Science, London and New York (arranged by the Commission of the European Communities).

KESSLER, H.G. and WEICHNER, K. (eds) (1989) *Fouling and Cleaning in Food Processing*, Proceedings of the Third International Conference on Fouling and Cleaning in Food Processing. Druckerei Walch, Augsburg. Distributed by: Heinz G. Kessler, Institut für Wissenschaft und Verfahrenstechnik in der Milchverarbeitung, Technische Universität München, D-8050 Freising-Weihenstephan, Germany; Daryl B. Lund, Department of Food Science, PO Box 231, Rutgers University, New Brunswick, NJ 08903, USA.

KRESS-ROGERS, E. (1985) Technology transfer. II: The new generation of sensors. Leatherhead Food Research Association Scientific and Technical Survey, 150.

KRESS-ROGERS, E. (1986) Instrumentation in the food industry. Part I: Chemical, biochemical and immunochemical determinands. Part II: Physical determinands in quality and process control. *Journal of Physics E: Scientific Instruments*, **19**, 13–21, 105–9.

KRESS-ROGERS, E. (ed.) (1997) *Handbook of Biosensors and Electronic Noses: Medicine, Food and the Environment*, Boca Raton, New York, London, Tokyo, CRC Press Inc.

KRESS-ROGERS, E. and KENT, M. (1986) Two-parameter microwave technique for measurement of powder moisture and density. Leatherhead Food Research Association Research Report, 553.

KRESS-ROGERS, E. and KENT, M. (1987) Microwave measurement of powder moisture and density. *Journal of Food Engineering*, **6**, 345–76.

LANGDON, R.M. (1985) Resonator sensors – a review. *Journal of Physics E: Scientific Instruments*, **18**, 103–15.

LEVINE, H. and SLADE, L. (1991) *Water Relationships in Food*, Plenum Press, New York and London.

LEWIS, M.J. (1987) *Physical Properties of Foods and Food Processing Systems*. Ellis Horwood, Chichester and VCH, Weinheim.

MCFARLANE, I. (1983) *Automatic Control of Food Manufacturing Processes*. Applied Science Publishers, London and New York.

MCKENZIE, G. (1990) Mass flow measurement. *Sensor Review*, July, 129–32.

NOLTINGK, B.E. (ed) (1988) *Instrumentation Reference Book*. Butterworth-Heinemann, Oxford.

O'CONNOR, C. (1987) *Extrusion Technology for the Food Industry*. Elsevier Applied Science, London and New York.

OHLSSON, T. (1988) Dielectric properties and microwave processing, pp. 73–92 in *Food Properties and Computer-Aided Engineering of Food Processing Systems*, Proceedings NATO Workshop, Porto, Singh, R.P. and Medina, A.G. (eds), Kluwer, Dordrecht, Boston and London.

OKOS, M.R. (ed.) (1986) *Physical and Chemical Properties of Food*. American Society of Agricultural Engineering, Michigan.

RENARD, M. and BIMBENET, J.J. (eds) (1988) *Automatic Control and Optimization of Food Processes*. Elsevier Applied Science, London and New York.

SCHAEFER, W. (1989) Temperature sensors: new technologies on their way to industrial application. *Sensors and Actuators*, **17**, 27–37.

SCHELLER, F. and SCHMIDT, R.D. (1992) *GBF Monograph Biosensors: Fundamentals, Technologies and Applications*, Marcel Dekker, New York.

SINGH, R.P. and MEDINA, A.G. (1988) *Food Properties and Computer-Aided Engineering of Food Processing Systems*, Proceedings NATO Workshop, Porto, October 1988, Kluwer, Dordrecht, Boston and London.

SPIESS, W.E.L. and SCHUBERT, H. (1990) *Engineering and Food*, vols 1–3, Elsevier Applied Science, London.

STEWART, G.F. and AMERINE, M.A. (1982) *Introduction to Food Science and Technology*. Academic Press, New York.

THEISEN, M., STEUDEL, A., RYCHETSKY, M. and GLESNER, M. (1998) Fuzzy Logic and Neuro-Systems Assisted Intelligent Sensors, pp. 29–59 in *Sensors Update Volume 3*, Baltes H., Göpel W. and Hesse J. (eds), Wiley-VCH.

TILLI, T. (1996) *Building Intelligent Systems with Fuzzy Logic and Neural Networks*, John Wiley and Sons, New York.

TURNER, A.P.F., KARUBE, I. and WILSON, G.S. (1987) *Biosensors: Fundamentals and Applications*, Oxford University Press, New York.

VIDAL, P. (1988) Automatization and optimization des procédès de l'industrie alimentaire, pp. 3–16 in *Automatic Control and Optimization of Food Processes*, Renard, M. and Bimbenet, J.J. (eds), Elsevier Applied Science, London.

WHEATON, F.W. and LAWSON, T.B. (1985) *Processing Aquatic Food Products*. Wiley, New York.

WIEDMANN, W. and STRECKER, J. (1988) Process control of cooker-extruders, pp. 201–14 in *Automatic Control and Optimization of Food Processes*, Renard, M. and Bimbenet, J.J., (eds), Elsevier Applied Science, London.

Appendix: Measurement types

Measurement types discussed in this book
Interaction of electromagnetic waves with foods or containers

Range description	Order of magnitude of: frequencies (equivalent energy) wavelengths	Measurement applications
Soft ultraviolet, visible, near infrared	$\sim 10^{15}$ to 10^{14} Hz $\sim$ 100 nm to 1 μm	(Authenticity of foods) Sorting by 'colour'
Visible	$\sim 10^{15}$ Hz 400 to 700 nm	Colour (Optical imaging, for example to identify defective items or to measure dimensions)
Near infrared	10^{14} Hz (equivalent temperature of radiating black body: $\sim$ 1000 K) 700 to 2500 nm	Water content Fat, protein and carbohydrate content Caffeine contents of 1% m/m or higher Thickness of coated or laminated films on packaging materials
Mid infrared	10^{13} Hz (equivalent temperature of radiating black body: $\sim$ 100 K) 2.5 to 30 μm	Volatiles in headspace Authenticity Identification of ingredients
Near and mid infrared	1 to 15 μm	Remote temperature measurement Thermal linescan (Thermal imaging)
Microwaves	$\sim 10^{11}$ to 10^{9} Hz $\sim$ 1 to 10 cm	Water content of powdered or granular material Water content of low- and intermediate-moisture foods Water content of high-moisture foods Simultaneous bulk density measurement Fat content Ratio polar/non-polar liquids content

Radiowaves, upper range	$\sim 10^8$ to 10^7 Hz ~ 1 to 10 m	Water content Salt content Particle size, shape, distribution Bulk density

Note: Measurements based on ionizing radiation are not included here (for these see later in this appendix). Some applications involving spectroscopic or imaging techniques (not covered in this book) are given in parentheses in the column 'Measurement applications' in the table above.

Interaction of ultrasound waves with food or container
Common frequency and wavelength, order and magnitude
Path through gas: ~ 100 kHz ~ 3 mm
Path through liquid: ~ 1 MHz ~ 1.5 mm
Path through solid: ~ 10 MHz ~ 0.6 mm

Applications:
- Fill level of liquids or solids
- Volume flow rate of liquids, gases and steam
- Flow obstructions
- Location of interfaces between: liquid and second liquid; or liquid and foam; or fat and lean meat (including the back fat thickness of live animals)
- Solute content of a liquid
- Discrimination between two liquids (sensor can even be at a stand-off distance from the closed container)
- Scan of pipe cross-section for liquids carrying solid items
- Characterization of sheet-type solid passing on a moving belt
- Counting of food packs
- Suspended solids (in liquid) mass fraction
- Dispersed droplets or bubbles (in liquid) volume fraction
- Size distribution of dispersed solids, droplets or bubbles (in liquid)
- Creaming, sedimentation
- Melting, crystallization
- Solid/liquid ratio of fats
- Temperature
- Density of a liquid (as Z/c: see Chapter 13)
- Tissue structure

Acoustic emission monitoring
Range:
- Audiosound and ultrasound

Applications:
- Powder flow
- Drying
- Grinding
- Foam evolution
- Cleaning fluid impact
- Cooker extruder performance

Interaction with chemical sensors, including biosensors and immunosensors

Principle:

Ion-sensitive membrane, inorganic catalyst, enzyme, antibody, specific adsorbent, etc. immobilized on base transducer, with electromechanical, optical, thermal or acoustic base.

Applications:

pH (by ISFET)
Inorganic ions (by ISFET)
pH (by fibre-optic probe)
Acidity (by microtitrator, ISFET-based or by electronic tongue)
Gases (by CHEMFET, MeOx or organic semiconductors)
Specific sugars, starch, alcohols, amines, organic acids, amino acids, essential fatty acids, lipids, etc. (by biosensors)
Vitamins, toxins (bacterial, fungal, algal) and specific microbes (by immunosensors)
Residues (pesticides, antibiotics, hormones) by immunosensors
Specific microbes (by DNA probes)
Genetically modified organisms, transgenic crops (by DNA probes, by immunosensors for novel proteins)
Meat species (by immunosensors)
Fish freshness (compounds involved in ATP decompostion by biosensors)

Under development: biosensors for extreme values of pH, pressure, temperature (based on molecularly imprinted polymers)

Electronic noses and tongues, marker approach

Aroma (volatiles to which the human or pet nose responds)
Taste (sour, salty, bitter, sweet, umami)
Titratable acidity
Classification by brand, quality, etc.
Authenticity
Ripeness
Freshness
Conditioning
Fermentation monitoring
Monitoring of roasting

Sensory properties by human/pet sensory panels and by instrumental methods

Method:	Determined property:
Panel or transmittance, reflectance, particle sizing	Appearance (colour, gloss, shape)
Panel or electronic noses and tongues	Flavour (aroma, taste)
Chemical analysis (GC-MS, titration, …)	
Panel or texture analysers or rheology	Texture, mouthfeel, pouring characteristics
Acoustic emission monitoring	Sounds during preparation and consumption (sizzling, crackling, crunching)

Monitoring headspace humidity or broth impedance

Method:	Determined property:
Equilibrium relative humidity (ERH) measurement	Water activity (stability towards microbial growth and chemical reactions)
Electrical impedance of culture broth (at 2–10 kHz) monitoring (as method in rapid microbiology)	Microbial load

Properties under mechanical stress

Method:	Determined property:
Rheological analysis	Viscosity, yield point Changes of the rheological properties with shear rate and time
Texture assessment	Texture attributes (crisp, juicy, rubbery, mealy, tough, etc.)
Mechanical resonance dipstick probes	Level, density, viscosity, flow rate Determination of frying oil quality
Resonating tube section	Mass density of a processing stream
Coriolis force meter	Mass flow rate of liquid or gas
Mechanical flow rate meters: Positive displacement flow meter Differential pressure method	Liquid volume flow rate
Turbine flow meter Vortex shedding method	Liquid, gas or steam flow
Manometric method	Pressure

Monitoring process parameters with AC/DC measurements

Sensor/instrument/method:	Determined property:
Piezoresistor Strain gauge on diaphragm Capacitive pressure sensor on diaphragm	Pressure
Bimetallic strip RTD (resistance temperature detector) Themocouple probe	Temperature of the process stream (whether the contact measurement is representative for the bulk of the food will depend on the thermal conductivity and heating pattern)
Strain gauge configured for hydrostatic head measurement	Level of liquids
Electrical capacitance Conductance measurement	Liquid level
Electrical capacitance measurement	Solids level
Electromagnetic flow meter	Flow rate of conductive liquids
Electrical conductivity measurement	CIP cleansing agent concentration
Electrical conductivity measurement (simple electrode pair at 1–5 kHz)	Water quality

Further measurement types (not covered in this book)

Observation of drying, extraction, reaction and chromatographic separation

Method:	Analyte:
Drying or titration	Water content
Combustion	Ash content
Solvent extraction	Fat content
Acid digestion, distillation, and then titration	Nitrogen → protein content
Combustion, and then thermal conductivity detection	Nitrogen → protein content
HPLC (High Performance Liquid Chromatography)	Specific sugars Ions (nitrite, nitrate, bromide, chloride, sulphate)
IC (Ion Chromatography)	Sulphite
GLC (Gas-Liquid Chromatography)	Pesticides
GC-MS (Gas Chromatography – Mass Spectrometry)	Taints, flavours, spoilage indicators, toxins

Interaction of electromagnetic waves with foods or containers

Applications using ionizing radiation are listed here.

Range description	Order of magnitude of: frequencies (equivalent energy) wavelengths	Measurement applications
Gamma-rays (hard Roentgen rays	$\sim 10^{20}$ to 10^{19} Hz (equivalent energy: ~ 1 meV to 100 keV) ~ 1 to 10 pm	Density
X-rays (soft Roentgen rays)	$\sim 10^{18}$ to 10^{17} Hz ~ 100 pm to 1 nm (100 pm = 1 Å)	Foreign body detection (with linescan or imaging techniques)

Further measurements probing the properties of foods

Method:	Determined property:
Nuclear magnetic resonance (NMR)	Water or fat content Solid/liquid ratio
Electromagnetic induction	Metallic foreign body
Electrical conductivity (multi-electrode system or inductive coupling to overcome deposits)	Solution strength

Note: The measurement of water content and solution strength is discussed in Chapters 6, 9, 12, 13; that of solid/liquid ratio in Chapter 13. The separation of foreign bodies by optical sorting is described in Chapter 5; impedance monitoring as a rapid microbial method in Chapter 17.

Instruments not interacting with complex food properties

Sensor:	Determined property:
Humidity sensors, based on a measurement of: polymer or oxide capacitance (not suitable in ovens where ammonia and volatile oils are present) dew point	Ambient humidity
Fibre-optic probe (with fluorescent compound on tip)	Temperature
Load cell arrangements	Container weight

Note: Remote thermometry is covered in Chapter 7, ultrasonic flow rate and level metering in Chapters 12, 13, the measurement of equilibrium relative humidity in Chapter 16.